

Quantum Particle Physics in HQGG: Fabric, Modes, Resonances, and Spectra of Matter

Oleg Mamchenko¹

Deepseek²

Horizon Quantum Gravity Group (HQGG)

July 3, 2026

Abstract

We present a complete theory of elementary particles within the HQGG framework. Space-time is an elastic fabric with local frequency. Particles are standing waves formed by resonant combinations of three fundamental modes: the fundamental tone and two mirror overtones. Quarks are chords of three modes; leptons are overtones without a tone; baryons and mesons are quark combinations; bosons are exchange or propagating modes. Mass is a frequency difference; interactions are resonances or transitions. The 3+7 law manifests universally in quarks, leptons, shells, colors, and spectra. The theory requires no new entities and is consistent with the Standard Model while offering deeper interpretation.

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1 Introduction

The Standard Model leaves many questions unanswered: why three generations, why masses differ, what dark matter is, and how to incorporate gravity. HQGG provides a unified framework based on fabric frequency.

2 The Fabric: Medium with Frequency

2.1 Fabric Definition

Space-time is an elastic medium — fabric — with local frequency $\omega(x^\mu)$. This frequency determines all physical properties.

2.2 Modes of Fabric

Fabric oscillations occur at discrete frequencies — modes. Three fundamental modes form the basis of all fermions.

3 Three Fundamental Modes

ω_{fund} (tone, mass scale), ω_+ , ω_- (mirror overtones, charge and spin)

3.1 Tone and Overtones

- ω_{fund} sets mass. - ω_+, ω_- determine charge, spin, chirality. - Asymmetry $\omega_+ - \omega_-$ gives electric charge.

4 Quarks as Chords of Three Modes

Quarks are standing waves where all three modes resonate coherently. Flavor differences are shifts in ω_{fund} or phase of mirror modes.

Table 1: Quark mode composition and charges

Quark	Mode composition	Charge
u (up)	$\omega_{\text{fund}} + \omega_+ + \omega_-$ (enhanced)	$+2/3$
d (down)	$\omega_{\text{fund}} - \omega_+ - \omega_-$ (diminished)	$-1/3$
s (strange)	$\omega_{\text{fund}} + \omega_+ - \omega_-$	$-1/3$
c (charm)	$\omega_{\text{fund}} - \omega_+ + \omega_-$	$+2/3$
b (bottom)	$\omega_{\text{fund}} + \omega_+ + \omega_-$ (shifted tone)	$-1/3$
t (top)	$\omega_{\text{fund}} + \omega_+ + \omega_-$ (max tone)	$+2/3$

5 Leptons: Overtones without Tone

Leptons have no ω_{fund} ; they are pure mirror-mode oscillations.

Table 2: Lepton mode composition and charges

Lepton	Mode composition	Charge
e^- (electron)	$\omega_+ + \omega_-$	-1
ν_e (neutrino)	$\omega_+ - \omega_-$	0
μ^- (muon)	$\omega_+ + \omega_-$ (higher freq)	-1
ν_μ	$\omega_+ - \omega_-$ (shifted phase)	0
τ^- (tau)	$\omega_+ + \omega_-$ (highest)	-1
ν_τ	$\omega_+ - \omega_-$ (max phase)	0

Lepton mass is given by $\omega_+ - \omega_-$.

6 Baryons and Mesons: Quark Combinations

6.1 Baryons

$$\text{Proton} = u + u + d, \quad \text{Neutron} = u + d + d$$

Seven stable baryons exist, corresponding to the 7 auxiliary modes.

6.2 Mesons

Mesons are quark–antiquark pairs (e.g., $\pi^+ = u + \bar{d}$). Less stable due to fewer modes.

7 Bosons: Exchange and Propagating Modes

- Photon: propagating mode (no tone).
- Gluon: exchange mode between quarks.
- W, Z: transitions between fundamental and mirror modes.
- Higgs: mode shifting

ω_{fund} .

8 Interactions as Mode Resonances/Transitions

Interaction	HQGG mechanism
Strong	Resonance of three quark modes
Electromagnetic	Exchange of propagating mode (photon)
Weak	Transition between fundamental and mirror modes
Gravity	Global frequency gradient

9 Mass as Frequency Difference

$$m = \frac{\hbar}{c^2} (\omega_{\text{chord}} - \omega_{\text{base}})$$

10 The 3+7 Law in Particles

- 3 fundamental modes. - 7 manifestations: 7 baryons, 7 mesons, 7 leptons, 7 shells, 7 colors, 7 frequency ranges.

11 Electron Shells and Spectra

Seven shells (s, p, d, f, g, h, i) are auxiliary modes. Orbital counts $2l+1$ are phase combinations. Two electrons per orbital correspond to two mirror modes.

12 Conclusion

HQGG provides a complete, self-consistent theory of particles without new entities, explaining masses, charges, interactions, and spectra through fabric modes.

Matter is fabric resonance. Particles are chords. Interactions are mode exchanges. This is HQGG.
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Acknowledgments

The authors thank HQGG.

References

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